Theory of Materialization in the Structural-Physical Framework

Source-faithful program manuscript

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Abstract

This manuscript is a source-faithful reconstruction of the first theory block of the materialization program. It separates imported book-level machinery, program-specific definitions, derived propositions, target theorems, and failure modes for the initial materialization regime.

The manuscript does not claim ex nihilo creation, full theorem closure of downstream engineering or prototype layers, or closure of later fabrication, surface, repair, device, adaptive, proto-agent, or self-fabricating regimes. Any downstream computational artifacts are recorded, when needed, only as non-evidential provenance and not as theorem evidence.

The present pass is limited to the first source-stable architecture of the program and to the first explicit specialization: crystalline and monatomic hard materialization as a target theorem chain.

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1 Protocol, Boundary, and Status Policy

The materialization program is a separate satellite theory. It is not an appendix to chemistry, and it is not a claim that matter appears from nowhere. It imports the structural, observer, composite, and material-class machinery of the main book, then adds a specification-and-assembly formalism with explicit defect, persistence, and failure boundaries.

The purpose of the present manuscript is narrow: to reconstruct the first source-stable theory block of the project in clean scientific English while keeping the dependency ladder explicit. Later crystalline, fabrication, surface, repair, and fabricator blocks remain downstream extensions rather than claims already closed here.

Boundary Principle M0. A materialized object is not primitive. It is a late observer-level realization of a deeper structural history. Accordingly, materialization is not "matter appearing from nowhere," but the successful production of a stable or metastable composite realization from precursor sectors under an admissible assembly history.

1.1 Status policy

This manuscript adopts the following status discipline:

- **[A] Definition/Axiom:** adopted or program-level primitive language.
- **[D] Derived Proposition/Theorem:** consequence expected from imported machinery plus program-specific definitions.
- **[T] Target Theorem:** intended closure statement not yet claimed at full rigor.
- **[F] Failure Mode:** mandatory audited way in which the program can fail.

The project remains rigorous only if successful cases, open closures, failure modes, and imported assumptions are stated on the same footing.

2 Imported Foundations and Boundary

The project studies materialization as a controlled passage from precursor structural resources to durable observer-level objects. It is not a replacement for the structural-physical ladder of the book. It is a satellite theory that depends on that ladder and asks how one reaches a target object once the relevant precursor sectors and control channels are available.

This separation is methodologically necessary. The book is organized as a dependency chain in which later regimes arise as observer-level shadows of deeper structural dynamics, and special domains are meant to be embedded into that chain rather than treated as independent primitives.

2.1 Imported foundations

The manuscript imports four already prepared layers.

First, the *structural level*: symbolic structures, sequential objects, h-patterns, EFU decomposition, transfer kernels, marked observables, and spectral control. The overall architecture of the book rests on the rule that structural organization precedes both semantic collapse and ordinary observables.

Second, the *observer level*: the book separates the observer as a projection/decategorification mechanism rather than a primitive thing inside space. Space, time, and ordinary observables are treated as derived from deeper causal histories.

Third, the *composite and bound-state level*: chemistry is formulated there as a theory of stable and metastable composite sectors. Atomic and molecular sectors arise through bound-state language, reaction barriers, metastable graphs, and material classes.

Fourth, the *many-body/material level*: material classes are already defined as stable many-body phases characterized by composition pattern, bond structure, symmetry or topology, gap structure, and coarse response. Crystalline, liquid, glass, polymeric, and electronic classes are explicitly separated there.

2.2 Standing imported assumptions

The present manuscript uses the following imported assumptions throughout:

H-Imp1. Observer projection. Observer-level access is given by the imported projection and decategorification machinery.

H-Imp2. Composite-sector support. Stable and metastable composite sectors are available as admissible physical carriers.

H-Imp3. Barrier language. Reaction barriers and metastable transition channels are available in the imported chemistry layer.

H-Imp4. Material classes. Stable and metastable many-body material classes are available, including the crystalline and glass classes.

H-Imp5. Crystalline recognition criterion. The crystalline class is recognized by long-range positional order or nonzero Bragg-type amplitudes.

H-Imp6. Chemical memory criterion. The chemical memory criterion is available when invoked.

These assumptions are not reproved in the present manuscript pass. They are imported from the main theory corpus and serve as standing background for later derived propositions and target theorems.

2.3 Explicit boundary

The project does not begin with a ready-made particle, field, atom, molecule, or ordinary macroscopic body as primitive. Fundamental theory begins with symbolic structures, admissible transformations, and later layers of composition, kernels, and observer-level shadows. Materialization therefore means successful realization from precursor resources under admissible assembly histories, not primitive material being posited at the start.

3 Specification Theory

Here begins the first genuinely new block of the materialization project. The central shift is that an ordinary thing is not identified by raw substance alone, but by passing a structured observer-relative specification.

Notation convention. Throughout the manuscript, the notation

$$X \models_O \Sigma_O$$

means that the realized sector X satisfies the observer-level predicate family of Σ_O within the tolerated defect budget and throughout the required persistence window. Likewise,

$$X \sim_{\Sigma_O} Y$$

denotes observer-equivalence relative to the same specification.

Definition 3.1 [A]. M1. Observer-level specification. Let $O = (\Pi_O, \Theta_O, \chi_O)$ be an observer in the imported sense. An *observer-level specification* is a triple

$$\Sigma_O = (\mathcal{P}_O, \varepsilon_O, \tau_O),$$

where $\mathcal{P}_O$ is a family of observer predicates or tests, ε_O is a tolerated defect budget, and τ_O is the required persistence window.

The first component says what counts as the same object for that observer. The second says how much deviation is admissible. The third says how long the object must remain valid. Once physical objects are observer-level shadows, the target of materialization cannot be given purely at the raw structural level; it must be stated as a set of acceptable observer-level properties.

Definition 3.2 [A]. M2. Predicate family. A predicate family $\mathcal{P}_O$ may include

$$\mathcal{P}_O = \mathcal{P}_{\text{geom}} \cup \mathcal{P}_{\text{comp}} \cup \mathcal{P}_{\text{mech}} \cup \mathcal{P}_{\text{therm}} \cup \mathcal{P}_{\text{elec}} \cup \mathcal{P}_{\text{spec}} \cup \mathcal{P}_{\text{func}}.$$

Respectively, these correspond to geometry, composition, mechanical response, thermal response, electrical response, spectral signature, and functional behavior.

This matches the imported notion of material class, which is characterized by structure, symmetry or topology, gap structure, and coarse mechanical, electrical, or thermal response.

Definition 3.3 [A]. M3. Defect budget. A *defect budget* ε_O is a rule assigning allowable deviations to the tests in $\mathcal{P}_O$. In the simplest case,

$$\varepsilon_O = (\varepsilon_{\text{geom}}, \varepsilon_{\text{comp}}, \varepsilon_{\text{spec}}, \varepsilon_{\text{func}}).$$

Thus a materialized object need not be exact in every microscopic detail. It must be acceptable relative to the observer's admissible tolerances.

Definition 3.4 [A]. M4. Persistence window. A *persistence window* τ_O is the minimal observer-time interval during which the target predicates remain valid. For long-lived material objects, one asks for

$$\tau_{\text{life}}(X) \gg \tau_O.$$

This is directly analogous to the book's metastability criteria, such as the glass criterion $\tau_{\text{relax}} \gg \tau_{\text{obs}}$ and the chemical memory criterion $\tau_{\text{flip}} \gg \tau_{\text{read}}$. Here $\tau_{\text{life}}(X)$ denotes the observer-level persistence surrogate of the realized object.

Definition 3.5 [A]. M5. Observer-equivalence of realized objects. Let X and Y be candidate realized objects. We say that they are *observer-equivalent relative to* Σ_O if

$$X \sim_{\Sigma_O} Y$$

whenever every predicate in $\mathcal{P}_O$ differs by no more than the allowed defect budget and both objects persist through the required time window.

This is the first answer to the identity problem: the theory should not ask whether two objects are literally the same microscopic assembly, but whether they are the same materialized object for the observer and purpose at hand.

4 Precursor and Resource Theory

Now we define what a materialization process is allowed to use.

Definition 4.1 [A]. M6. Precursor reservoir. A *precursor reservoir* is a specified set of admissible structural, composite, chemical, or many-body sectors available for assembly.

Materialization is always relative to available precursor resources. The theory does not allow unrestricted ex nihilo creation.

Axiom 4.2 [A]. M-A1. No ex nihilo materialization. If no admissible precursor reservoir is specified, then the term materialization is undefined.

This is the correct project-level replacement for magical language. The best the theory can ever do is convert available resources into a new stable observer-level object.

[Open-Calibration] Resource profile remark (M7). Later extensions of the materialization program may equip a precursor reservoir with a resource profile

$$\mathcal{R}(R_{\mathrm{pre}}) = (\mathcal{M}, \mathcal{Q}, \mathcal{I}, \mathcal{S}, \mathcal{B}),$$

recording matter-like resources, conserved charges, informational content, structural complexity, and accessible barrier structure. This bookkeeping is not used in Parts I–XI of the present manuscript pass.

5 Assembly Histories and Controlled Channels

The next genuinely new object is the materialization channel.

Definition 5.1 [A]. M8. Assembly history. An *assembly history* is an admissible causal history

$$\Gamma : R_{\mathrm{pre}} \rightsquigarrow X$$

that transforms the precursor reservoir into a candidate target sector X.

Because the imported framework already treats physical histories as causal rewriting histories rather than ready-made motion in a background space, this is the natural form of a materialization process.

Definition 5.2 [A]. M9. Controlled assembly channel. A *controlled assembly channel* is a family of assembly histories

$$\mathfrak{C} = \{\Gamma_u\}_{u \in \mathcal{U}}$$

parametrized by control data u, such as fields, catalysts, geometric confinement, temperature programs, boundary conditions, or informational templates.

This is where the project becomes more than chemistry. Reaction graphs and catalysts already exist in the imported chemistry block, but they are now subordinated to a new question: how to steer histories toward a desired observer-level thing.

Definition 5.3 [A]. M10. Target-reaching channel. A controlled channel $\mathfrak{C}$ is *target-reaching* for specification Σ_O if there exists a control u and a history Γ_u such that the resulting sector X_u satisfies

$$X_u \models_O \Sigma_O.$$

This is the first formal bridge from precursor dynamics to the statement that the target object was obtained.

6 Success Criteria: Soft and Hard Materialization

Now the manuscript can define success itself.

Definition 6.1 [A]. M11. Soft materialization. A channel realizes *soft materialization* if it produces an observer-valid image while active, but the realized sector ceases to satisfy the specification after control removal.

Formally, X_u satisfies Σ_O only under maintained control conditions. This corresponds to field-supported, confinement-supported, or otherwise externally stabilized objects.

Definition 6.2 [A]. M12. Hard materialization. A channel realizes *hard materialization* if after control removal the resulting sector X:

1. remains observer-distinguishable,
2. remains inside the defect budget,
3. persists through the required window, and
4. belongs to a stable or metastable material class.

The last clause is imported directly from the chemistry/material block: the theory already recognizes stable and metastable material classes as the correct coarse carriers of durable physical organization.

Definition 6.3 [A]. M13. Successful materialization. A realized sector X counts as successfully materialized for observer O and specification Σ_O if

$$X \models_O \Sigma_O$$

in either the soft or the hard sense.

Soft success means observer-valid realization under maintained control. Hard materialization is the stricter case in which observer-validity persists after control removal through the required window and remains supported by a stable or metastable material class.

7 Hierarchy Placement and First Derived Propositions

The present source pass does not yet build the full hierarchy $\mathcal{H}_0$ through $\mathcal{H}_8$ in theorem detail. That hierarchy already exists as an internal program ledger and is reproduced in Appendix A. Within that ledger, the first successful materialized object belongs to the lowest operational level $\mathcal{H}_0$: hard materialization of a simple crystalline body.

For the current manuscript, the hierarchy therefore plays a boundary-setting role rather than a new closure claim. The first derived propositions of the source belong exactly at this transition point between the basic language of Parts II–V and the future hierarchy ladder.

Proposition 7.1 [D]. M-P1. Specification-relativity. *One and the same underlying structural sector may satisfy one observer-level specification and fail another.*

This follows immediately because the imported framework already makes the observer a triple of projection and decategorification maps rather than an irrelevant external spectator.

Imported support: H-Imp1.

Proposition 7.2 [D]. M-P2. Materialization is stronger than synthesis. *Not every synthesized sector is materialized. A synthesized sector counts as materialized only if it survives the observer-level validation map with controlled defects and sufficient persistence.*

This marks the point at which the materialization project diverges from the chemistry chapter. Chemistry gives stable and metastable sectors; materialization adds the target-specification layer.

Imported support: H-Imp1–H-Imp4.

Proposition 7.3 [D]. M-P3. Information-bearing sectors are privileged precursors. *If a precursor system contains informational chemical structures with long readout time and well-separated metastable states, then it is a better candidate reservoir for controlled materialization than a reservoir with the same raw composition but no stable record capacity.*

This proposition is suggested by the imported chemical memory criterion, where metastability and observer-distinguishability already function as primitive physical memory.

Imported support: H-Imp2, H-Imp4, H-Imp6.

8 First Target Theorems

The first target theorems of the project can now be stated without pretending that they are already closed at full rigor.

Target Theorem 8.1 [T]. M-T1. No ex nihilo theorem. *No successful materialization theorem can be proved without a nonempty precursor reservoir and a valid assembly history.*

Imported support: H-Imp1–H-Imp3.

Target Theorem 8.2 [T]. M-T2. Soft/hard separation theorem. *There exist channels that satisfy the observer-level specification only under maintained control, and other channels whose output survives as a metastable or stable material class after control removal.*

Imported support: H-Imp1–H-Imp4.

Target Theorem 8.3 [T]. M-T3. Observer-relative identity theorem. *Two different assembly histories may produce non-isomorphic microscopic sectors that are nevertheless the same materialized object relative to Σ_O.*

Imported support: H-Imp1.

Target Theorem 8.4 [T]. M-T4. Minimal precursor theorem. *For a fixed specification Σ_O, there exists a partial order on precursor reservoirs by sufficiency, and one may ask for minimal sufficient reservoirs.*

Imported support: H-Imp2–H-Imp3.

This last theorem is especially important if the program later grows into an explicit fabricator theory.

The formal-support provenance table for these target statements is recorded in Appendix A.

9 Failure Modes and First Model Problem

The project must already be stricter than a slogan. Its failure modes are part of the theory, not an embarrassment to be deferred.

Failure Mode 9.1 [F]. M-F1. No stable branch. The produced sector has no stable pole or is too broad a resonance.

Failure Mode 9.2 [F]. M-F2. No coarse observability. The sector exists microscopically but does not survive observer coarse-graining as a distinct object.

Failure Mode 9.3 [F]. M-F3. No persistence. The sector satisfies the predicates only transiently, for less than τ_O.

Failure Mode 9.4 [F]. M-F4. Defect overflow. The realized object violates the defect budget.

Failure Mode 9.5 [F]. M-F5. No material-class support. The channel produces a transient assembly but not a stable or metastable material carrier.

These failure modes extend the chemistry chapter's own audited failure vocabulary: absence of atomic poles, overly broad molecular sectors, failure of slow-fast separation, lack of stable coarse observables, and ill-defined barriers.

9.1 The first concrete model problem

To avoid drowning in generality, the program needs a first partial arena. The source fixes it as follows:

> First model problem: hard materialization of a simple crystalline body.

This is the correct first object because the book already provides a crystalline class, many-body/material language, barriers, reactions, and metastability. For such an object it is simplest to define geometry, spectral signatures, defect budget, and persistence window.

So the next formal step is not a theory of every possible object, but specification and hard materialization of a crystalline target object.

10 Engineering Transition and Next Dependencies

The source is explicit about the next two turns of the project, and both are now represented in this manuscript. Parts I–IX establish the general language, Part X introduces explicit crystalline targets and defect calculus, and Part XI converts that crystalline target into an operational fabrication problem with seeds, growth histories, completion thresholds, and stopping criteria.

That transition is exactly where the theory becomes operational rather than merely axiomatic. The crystalline chapter identifies the first object class for which the full lower dependency chain can be stated explicitly. The monatomic chapter then turns that class into a controlled completion problem rather than a mere existence statement.

10.1 P0 bridge

Within the internal program ledger, $\mathcal{H}_0$ is the first candidate level for later engineering translation, once a separate calibration layer is fixed. A downstream P0 computational line may track threshold observables, crystalline quality surrogates, metastability windows, growth surrogates, and run-level verdict classes for the lowest hard-materialization regime. These downstream artifacts are not used here as theorem evidence. They are recorded separately in Appendix B as non-evidential provenance.

10.2 Formalization and computational dependencies

The first formalization targets are the definitions rendered in this draft: observer-level specification, precursor reservoir, assembly history, controlled channel, soft versus hard materialization, and successful materialization. Downstream computational observables may later normalize defect-budget surrogates, persistence surrogates, and crystalline quality functionals, but those calibrations are not part of the present core manuscript.

The present manuscript therefore closes with a strict boundary: Parts I–XI are reconstructed here, while later surface, repair, device, and stronger engineering regimes remain downstream work.

10.3 Companion computational and laboratory documents

The present manuscript remains the core theory document of the materialization program. A computational companion may operationalize the lowest regime in a simulator by introducing calibration packages, verdict rules, sweep protocols, and reproducibility contracts for the P0 line. A laboratory companion may translate the same regime into apparatus-level protocols, measured observables, acceptance criteria, and post-control validation rules.

Neither companion is used here as theorem evidence. Both belong to downstream calibration and realizability work for the lowest regime and must be read as operational continuations of the core manuscript rather than as closure of the theorem chain stated in Parts I–XI.

11 Crystalline Target Objects, Defect Calculus, and the First Hard-Materialization Theorem

The book already provides almost all of the background needed for the first explicit specialization. Chemistry is treated there as a theory of stable and metastable composite sectors, and atomic or molecular sectors count as physically admissible when their reduced resolvent has a stable isolated pole or a sufficiently narrow resonance branch. Material class is then defined as a stable many-body phase of atomic or molecular sectors, and the crystalline class is recognized by long-range positional order or nonzero Bragg-type amplitudes.

This is why the crystalline object is the first case in which the full lower dependency chain can be stated explicitly:

$$\begin{aligned}\text{precursor reservoir} &\to \text{assembly history} \to \text{bound composite sectors}\\ &\to \text{many-body crystalline phase} \to \text{observer-level object.}\end{aligned}$$

Definition 11.1 [A]. M14. Crystalline target specification. A crystalline specification for observer O is the quadruple

$$\Sigma_O^{\mathrm{cryst}} = (\Lambda_O, \mathcal{C}_O, \varepsilon_O, \tau_O),$$

where Λ_O is an observer-level lattice pattern, $\mathcal{C}_O$ is a composition profile, ε_O is a defect budget, and τ_O is the required persistence window.

This specification rests on the imported notion of material class: a material is identified not by bare substance alone, but by a package of structure, spectral regime, and coarse response. For the crystalline class, the key marker is long-range order or Bragg-type amplitudes.

Definition 11.2 [A]. M15. Observer crystalline predicate. Let $\rho_O(x)$ be the observer-level coarse density of a realized object x. The crystalline predicate is written

$$\mathrm{Cryst}_O(x) = 1$$

whenever at least one imported crystalline-recognition criterion holds, such as

$$\text{(i) long-range positional order,} \qquad \text{or} \qquad \text{(ii) nonzero Bragg-type amplitudes.}$$

This is not new physical content, but a transfer of the imported crystalline-recognition criterion into the language of the materialization project.

Imported support: H-Imp2–H-Imp5.

Definition 11.3 [A]. M16. Crystalline identity window. Two realized crystalline objects X and Y count as the same target object for O if

$$\Lambda_O(X) \approx \Lambda_O(Y), \qquad \mathcal{C}_O(X) \approx \mathcal{C}_O(Y),$$

their Bragg profiles differ by no more than ε_O, and both preserve this equivalence through the window τ_O.

11.1 Defect calculus

The project now needs a genuinely new tool: not merely the statement that defects exist, but a formal way to account for them inside materialization.

Definition 11.4 [A]. M17. Defect sector. Let X_{ideal} be an ideal crystalline target. The defect sector $D(X)$ denotes the structured deviation of the realized state X from X_{ideal} at observer and assembly level. No literal tensor factorization is assumed in the present manuscript pass. The term defect sector refers only to the organized deviation responsible for failure of exact target agreement without necessarily destroying the object itself.

Definition 11.5 [A]. M18. Defect observables. Introduce defect observables

$$\delta_{\mathrm{lat}}(X), \quad \delta_{\mathrm{comp}}(X), \quad \delta_{\mathrm{spec}}(X),$$

where these measure, respectively, deviation from the lattice pattern, deviation of composition, and broadening or smearing of spectral or Bragg signatures.

Definition 11.6 [A]. M19. Defect norm. The full observer-weighted defect norm is defined by

$$|D(X)|_O = w_{\mathrm{lat}}\,\delta_{\mathrm{lat}}(X) + w_{\mathrm{comp}}\,\delta_{\mathrm{comp}}(X) + w_{\mathrm{spec}}\,\delta_{\mathrm{spec}}(X),$$

where the weights depend on observer purpose.

Functional defect terms may appear in later usable-body or device layers, but they are not part of the present crystalline manuscript pass.

Definition 11.7 [A]. M20. Admissible defect budget. A crystalline object satisfies the defect budget if

$$|D(X)|_O \le \varepsilon_O.$$

The shift is fundamental: materialization does not demand ideality, only controllable nonideality.

11.2 Crystalline precursors and channels

Definition 11.8 [A]. M21. Crystalline precursor reservoir. A crystalline precursor reservoir is a reservoir

$$R_{\mathrm{pre}}^{\mathrm{cryst}}$$

consisting of atomic, molecular, ionic, cluster, or already-correlated sectors suitable for transition into a many-body phase with lattice order.

This definition leans directly on the chemistry chapter, where atomic and molecular sectors are built as bound composite branches and material classes as many-body phases of those sectors.

Definition 11.9 [A]. M22. Crystallization channel. A *crystallization channel* is a controlled assembly channel

$$\mathfrak{C}_{\mathrm{cryst}} = \{\Gamma_u\}_{u \in \mathcal{U}}$$

such that, for some control program u, the output sector X_u enters the crystalline class.

The control data may include temperature, field, pressure, a growth boundary, a template, a seed sector, a cooling schedule, a chemical gradient, or other admissible steering data.

Definition 11.10 [A]. M23. Seeded hard materialization. If the channel $\mathfrak{C}_{\mathrm{cryst}}$ uses a small seed sector $S_{\mathrm{seed}} \subset R_{\mathrm{pre}}^{\mathrm{cryst}}$ that fixes the lattice orientation or topology class of the growing body, the regime is called *seeded hard materialization*.

This is important because many nontrivial cases are easier to realize by controlled growth around a stable nucleus than by attempting creation from scratch.

11.3 Successful crystalline materialization

Definition 11.11 [A]. M24. Hard-materialized crystalline realization. A crystalline target X counts as hard-materialized for observer O if all of the following hold simultaneously:

$$\text{(a) } \mathrm{Cryst}_O(X) = 1,$$

$$\text{(b) } |D(X)|_O \le \varepsilon_O,$$

$$\text{(c) } \tau_{\mathrm{life}}(X) \ge \tau_O,$$

(d) X belongs to a stable or metastable material class,

(e) X descends from an admissible precursor reservoir through an admissible assembly history.

Clause (d) is imported from the chemistry and material layer; clause (e) is a genuinely program-internal materialization condition.

Imported support: H-Imp2–H-Imp5.

In the present manuscript pass, the defect budget and persistence window are observer-relative admissibility parameters rather than canonically normalized constants. The persistence surrogate $\tau_{\mathrm{life}}(X)$ is observer-relative and is not canonically normalized in the present manuscript pass.

Target Theorem 11.12 [T]. M-C1. Hard materialization theorem for a crystalline target. *Let a crystalline specification $\Sigma_O^{\mathrm{cryst}}$ and precursor reservoir $R_{\mathrm{pre}}^{\mathrm{cryst}}$ be given. Assume that:*

1. *imported composite-sector support is available for the relevant precursor branches;*
2. *an admissible target-reaching assembly history Γ_u from $R_{\mathrm{pre}}^{\mathrm{cryst}}$ to X_u is specified;*
3. *the observer-level crystalline predicate Cryst_O is calibrated for the present target class;*
4. *the defect budget ε_O is explicitly fixed; and*
5. *a post-control persistence criterion for the window τ_O is separately specified.*

Then the output sector X_u counts as a hard-materialized crystalline realization relative to the present manuscript-level criteria.

Imported support: H-Imp2–H-Imp5.

Dependency note (non-proof). No proof is claimed in the present manuscript pass. This paragraph records only the imported dependency chain supporting the target statement. The imported composite, chemistry, and material-class layers provide the background on stable or metastable sectors, reaction barriers, and crystalline recognition; the materialization project adds only the observer-relative specification and defect-budget bridge.

11.4 Soft and hard crystalline materialization

Definition 11.13 [A]. M25. Soft crystalline materialization. Soft crystalline materialization occurs if Bragg structure and observer-valid geometry persist only while the control channel remains active:

$$\Gamma_u \text{ active } \Rightarrow X_u \models_O \Sigma_O^{\text{cryst}},$$

but after removal of control,

$$X_u \not\models_O \Sigma_O^{\text{cryst}}.$$

This corresponds roughly to an object sustained only inside a trapping field, an external confinement regime, or under continuous energy injection.

Definition 11.14 [A]. M26. Hard crystalline materialization. Hard crystalline materialization occurs if, after the channel is completed, the object remains in a stable or metastable crystalline class and preserves Σ_O^{cryst} across the window τ_O.

11.5 Crystalline failure modes

Failure Mode 11.15 [F]. C-F1. Precursor insufficiency. The reservoir contains no sectors capable of reaching the required lattice or composition class.

Failure Mode 11.16 [F]. C-F2. No composite stabilization. Local bound branches do not arise, or they are too broad to support assembly. This is consistent with the chemistry layer, where atomic poles may be absent or molecular sectors too broad or short-lived.

Failure Mode 11.17 [F]. C-F3. Glass capture. The system falls into an amorphous metastable regime rather than the crystalline class. This is especially natural because the glass class is already separated from the crystalline class and is characterized by missing long-range order together with slow relaxation.

Failure Mode 11.18 [F]. C-F4. Defect overflow. The crystal forms, but $|D(X)|_O > \varepsilon_O$.

Failure Mode 11.19 [F]. C-F5. No coarse observability. The deeper structure exists, but for the given observer there is no stable Bragg or geometric signature.

Failure Mode 11.20 [F]. C-F6. Persistence failure. The object satisfies the target specification only for too short a time, less than τ_O.

11.6 The first crystalline toy problem

To keep the theory from remaining purely abstract, the source immediately fixes the next step as a narrow specialization.

Toy Problem C1. Build a theory of hard materialization for a monatomic crystalline body with a single-component precursor reservoir. One should then specify

$$\Sigma_O^{\text{mono}} = (\Lambda_{\text{mono}}, \mathcal{C}_{\text{mono}}, \varepsilon_O, \tau_O),$$

where $\mathcal{C}_{\text{mono}}$ requires a single chemical species and Λ_{mono} requires one admissible lattice class.

Inside this toy problem one can already introduce seed versus seedless growth, minimal reservoir size, defect scaling with growth time, and a criterion of successful completion. That is exactly why the source points next to the monatomic fabrication chapter.

12 Monatomic Fabrication and Completion Criteria

Part X already says what successful crystalline materialization is. The next source block asks a sharper question: how does one actually drive a crystalline body to completion? This requires an operational layer with seed sectors, growth histories, completion thresholds, defect recurrence, and stopping criteria.

The move is still source-faithful to the imported background. The chemistry layer already treats reactions as a graph of metastable sectors with barriers and transition channels, and catalyst sectors are defined precisely by lowering barriers without substantially changing the global free-energy difference. Material class is then a stable many-body phase, and the crystalline class is recognized by long-range order or nonzero Bragg-type amplitudes.

The next project step is therefore to formalize a procedure that brings such a crystal to the status of a completed object.

12.1 Specialization to the monatomic case

Definition 12.1 [A]. M27. Monatomic crystalline target. A monatomic crystalline target for observer O is the specification

$$\Sigma_O^{\mathrm{mono}} = (\Lambda_O^{(a)}, a, \varepsilon_O, \tau_O),$$

where $\Lambda_O^{(a)}$ is an admissible lattice class for one chemical species a, a is the unique allowed species, ε_O is the defect budget, and τ_O is the required persistence window.

In this regime the composition part simplifies substantially, so composition defect and lattice defect can be studied almost separately.

Definition 12.2 [A]. M28. Monatomic precursor reservoir. A monatomic precursor reservoir is a reservoir

$$R_{\mathrm{pre}}^{(a)}$$

consisting of available atomic, ionic, cluster, or already-bound monatomic sectors of the same species a.

This specialization is admissible precisely because the imported chemistry layer already allows atomic sectors, molecular sectors, bond classes, and material classes to be treated within a single language of composite regimes.

12.2 Seed sector

Definition 12.3 [A]. M29. Seed sector. A seed sector S_{seed} is a small stable or metastable composite sector that already carries:

1. a local monatomic composition class;
2. a local lattice pattern compatible with $\Lambda_O^{(a)}$; and
3. a sufficiently small defect norm.

So the seed is not an arbitrary piece of matter, but a minimal nucleus of the required structural regime.

Definition 12.4 [A]. M30. Admissible seed. A seed sector is admissible for Σ_O^{mono} if it contains only the allowed species a, exhibits local order compatible with the target lattice class $\Lambda_O^{(a)}$, and satisfies

$$|D(S_{\mathrm{seed}})|_O \leq \varepsilon_{\mathrm{seed}},$$

where $\varepsilon_{\mathrm{seed}}$ is a seed tolerance, typically stricter than the final defect budget.

Proposition 12.5 [D]. M-P4. Seed as a barrier-lowering object. *An admissible seed acts as a special control sector that lowers the effective barrier of crystalline assembly.*

This directly parallels the imported definition of a catalyst sector: a catalyst lowers a reaction barrier while leaving the overall free-energy difference almost unchanged. Within the present project, the seed is not merely a catalyst but a structural catalyst for the correct order class.

Imported support: H-Imp2–H-Imp5.

12.3 Growth histories

Definition 12.6 [A]. M31. Growth front. For a partially assembled object X_n, define the growth front

$$F(X_n)$$

as the set of observer-relevant boundary sites or boundary cells at which new precursor units can be attached without destroying the target lattice class.

Definition 12.7 [A]. M32. Elementary growth event. An elementary growth event is an admissible local history step

$$X_n + r \to X_{n+1}, \qquad r \in R_{\mathrm{pre}}^{(a)},$$

such that the new sector X_{n+1} enlarges the target object rather than merely rearranging the old part.

Definition 12.8 [A]. M33. Growth history. A growth history is a causal chain

$$\Gamma_{\mathrm{grow}} : S_{\mathrm{seed}} = X_0 \to X_1 \to \cdots \to X_N,$$

where each step is an elementary growth event and all intermediate states preserve monatomic compatibility.

Definition 12.9 [A]. M34. Ordered growth channel. A growth history is called ordered if there exists a control program u such that each intermediate state X_n remains locally compatible with the target lattice class outside the admissible defect envelope.

Materialization is not mere mass accumulation, but preservation of the correct order class along the whole growth history.

12.4 Completion functional

Definition 12.10 [A]. M35. Completion observables. For a partially realized object X, introduce the observables:

1. $V_{\mathrm{occ}}(X)$, the occupied target-volume fraction;
2. $Q_\Lambda(X)$, the lattice-order quality;
3. $|D(X)|_O$, the defect norm.

Definition 12.11 [A]. M36. Completion functional. Define the completion functional by

$$\mathcal{C}_O(X) = \Big(V_{\mathrm{occ}}(X), Q_\Lambda(X), |D(X)|_O\Big).$$

Definition 12.12 [A]. M37. Completed monatomic object. A partially assembled object X counts as completed for observer O if

$$V_{\mathrm{occ}}(X) \geq v^*, \qquad Q_\Lambda(X) \geq q^*, \qquad |D(X)|_O \leq \varepsilon_O.$$

Here v^* and q^* are observer-relative completion thresholds.

Completion is therefore a checkable attainment of threshold data.

In the present manuscript pass, the thresholds v^* and q^* are observer-relative completion thresholds. Their quantitative calibration is not claimed here.

12.5 Defect accumulation law

Definition 12.13 [A]. M38. Defect increment. At the nth growth step, let the defect norm change by

$$\Delta_D(n) = |D(X_{n+1})|_O - |D(X_n)|_O.$$

Standing observables for the ordered-growth regime. In the present manuscript pass, ξ_n denotes stochastic defect injection at step n, J_n denotes front mismatch at step n, λ denotes the retained fraction of inherited defect load, and η, ζ are sensitivity coefficients. These observables are not derived here; they are standing quantities of the monatomic operational layer for $n = 0, \ldots, N-1$.

Axiom 12.14 [A]. M-A7. Controlled defect recurrence. In a controlled growth channel, let the defect norm satisfy the recurrence estimate

$$|D(X_{n+1})|_O \leq \lambda |D(X_n)|_O + \eta\, \xi_n + \zeta\, J_n,$$

where $0 \leq \lambda \leq 1$. The recurrence allows inherited defects, stochastic injection, and front mismatch to be tracked within one operational estimate.

Proposition 12.15 [D]. M-P5. Bounded-defect growth criterion. *Under the standing observables introduced above, if*

$$\sup_n \xi_n < \infty, \qquad \sup_n J_n < \infty, \qquad \lambda < 1,$$

then the defect norm remains uniformly bounded along the growth history.

Controlled growth is possible only when defects do not run away.

Imported support: H-Imp2–H-Imp5.

12.6 Seeded completion theorem

Target Theorem 12.16 [T]. M-C2. Seeded hard-materialization for a monatomic crystal. *Let a specification Σ_O^{mono}, precursor reservoir $R_{\mathrm{pre}}^{(a)}$, admissible seed S_{seed}, and thresholds $(v_*, q_*, \varepsilon_O, \tau_O)$ be given. Assume that there exists a controlled ordered growth channel*

$$\Gamma_{\mathrm{grow}} : S_{\mathrm{seed}} \to X_N$$

such that:

1. *each growth event is compatible with the monatomic composition class;*
2. *admissible local bound branches are available throughout the ordered-growth regime;*
3. *the lattice-order quality remains compatible with the target lattice class throughout the ordered-growth regime;*
4. *the defect recurrence remains bounded;*
5. *the completion thresholds*

$$V_{\mathrm{occ}}(X_N) \geq v^*, \qquad Q_\Lambda(X_N) \geq q^*, \qquad |D(X_N)|_O \leq \varepsilon_O$$

   are reached; and
6. *a separately specified post-control persistence criterion for the window τ_O is satisfied.*

Then X_N counts as a hard-materialized monatomic crystalline realization for observer O relative to the present manuscript-level criteria.

The present manuscript pass does not fix a unique persistence surrogate. It records only the dependency that such a post-control criterion must be supplied for hard materialization.

Imported support: H-Imp2–H-Imp5.

Dependency note (non-proof). No proof is claimed in the present manuscript pass; this paragraph records only the imported dependency chain supporting the target statement. The imported chemistry and material-class layers supply stable local branches, metastable transition language, and crystalline recognition; the present project adds the seed sector, growth front, completion thresholds, and bounded-defect bridge.

No unique quantitative calibration of v^*, q^*, or the post-control persistence surrogate is claimed in the present manuscript pass. Accordingly, M-C2 remains a target theorem of the present manuscript pass rather than a closed theorem.

12.7 Seedless growth as a separate regime

Definition 12.17 [A]. M39. Seedless nucleation channel. A seedless nucleation channel is a controlled channel in which, instead of an externally supplied S_{seed}, there appears a spontaneous or induced nucleus

$$\varnothing \to S_{\mathrm{nuc}}$$

inside the precursor reservoir, after which the subsequent history proceeds as an ordinary growth channel.

Target Theorem 12.18 [T]. M-C3. Nucleation-to-seeded reduction. *If there exists a critical nucleus S_{nuc} satisfying the admissible-seed conditions, then the problem of seedless materialization reduces locally to seeded hard materialization.*

The point is methodological as well as technical: the theory should not rebuild every regime from scratch if a more complicated regime can be reduced to one that is already formalized.

Imported support: H-Imp2–H-Imp4.

12.8 Completion time

Definition 12.19 [A]. M40. Completion step. For target thresholds $(v_*, q_*, \varepsilon_O)$, define the completion step by

$$N_{\mathrm{comp}} = \min\{n : \mathcal{C}_O(X_n) \text{ satisfies the completion criteria}\},$$

whenever such a step exists.

Proposition 12.20 [D]. M-P6. Finite completion under positive net growth. *If ordered growth has positive net filling rate and bounded defects, then the completion step is finite:*

$$N_{\mathrm{comp}} < \infty.$$

This is an operational statement: the object can not only be described, but also driven to a finished state in finite time.

Imported support: H-Imp2–H-Imp5.

12.9 Failure modes for monatomic fabrication

Failure Mode 12.21 [F]. M-F6. No admissible seed. No small nucleus exists that is compatible with the target lattice class.

Failure Mode 12.22 [F]. M-F7. Front disordering. The mismatch current at the growth front becomes too large, and ordered growth cannot be maintained.

Failure Mode 12.23 [F]. M-F8. Defect runaway. The defect norm $|D(X_n)|_O$ ceases to remain bounded, and the object loses admissibility before completion.

Failure Mode 12.24 [F]. M-F9. Premature glass capture. The system becomes trapped in an amorphous metastable class rather than the crystalline class. This follows the imported distinction in which the glass class is marked by absent long-range crystalline order together with slow relaxation.

Failure Mode 12.25 [F]. M-F10. Threshold incompletion. The bulk lattice is already formed, but the occupied-volume and lattice-order thresholds do not jointly stabilize within the completion window, so the observer-level object does not count as completed.

12.10 First computational schema of the monatomic regime

The source reuses the label M-A1 here for the first computational schema of the monatomic subproject. To keep manuscript references unambiguous, the present draft records that item as M-A1-Mono while preserving the source intent.

[A]. M-A1-Mono. Monatomic computational schema. Input:

$$(\Sigma_O^{\mathrm{mono}}, R_{\mathrm{pre}}^{(a)}, u).$$

Steps:

1. find admissible seeds $S_{\mathrm{seed}} \subset R_{\mathrm{pre}}^{(a)}$;
2. evaluate front-compatible growth events;
3. compute transition barriers and an effective growth path;

4. evolve the defect recurrence;
5. evaluate the completion functional $\mathcal{C}_O(X_n)$;
6. stop at the first $n = N_{\mathrm{comp}}$ for which the completion criteria hold; and
7. separately verify persistence after control removal.

This item matters because the larger book insists on separating internal theory from the quantitative pipeline while also demanding explicit failure modes and an explicit computational recipe for major projects.

12.11 Section outcome

After Part XI, the materialization project has a distinct seed sector, a distinct growth front, a distinct completion functional, the first bounded-defect criterion, and the first nontrivial hard-materialization target theorem for a monatomic crystal.

At this point the project has acquired the minimal operational language needed for a theory of controlled completion. Full theorem closure remains downstream work.

13 What Is Not Claimed in This Manuscript

The present manuscript does not claim:

1. ex nihilo creation;
2. theorem closure of downstream computational or prototype lines;
3. closure of later surface, repair, usable-body, device, adaptive, proto-agent, or fabricator regimes;
4. quantitative calibration of all observables introduced here; and
5. full engineering closure of the materialization program.

The manuscript claims only a source-faithful reconstruction of the first theory block and the first explicit crystalline and monatomic target-theorem architecture.

A Provenance and Source Ledger of the Wider Materialization Program

This appendix records the wider project ledger used for provenance and scope control. Not every item listed below is rendered as a full statement in the present core manuscript.

Materialization Symbol and Status Glossary

This table normalizes the vocabulary of the materialization program across the main theory draft and the internal source ledger.

Status Labels

Label	Meaning
[A]	Definition or axiom of the materialization program.
[D]	Derived proposition or theorem expected from the imported framework and the program-specific definitions.
[T]	Target theorem not yet claimed as closed at full rigor.
[F]	Failure mode or audited failure statement of the materialization program.
[Internal-M]	Internal construction of the materialization project.
[Imported-Book]	Imported machinery from the main book.
[Internal→Observer]	Internal construction whose meaning is finalized only after observer-level calibration.
[Open-Calibration]	Conceptually fixed item whose numerical calibration remains open.
[Open-Closure]	Item with architecture present but theorem-chain closure still open.

Core Symbols and Objects

Symbol	Normalized meaning
Σ_O	Observer-relative specification of the target object.
$\Sigma_O^{\mathrm{cryst}}$	Crystalline observer-relative specification with lattice, composition, defect-budget, and persistence data.
Σ_O^{mono}	Monatomic crystalline specification with one lattice class, one species, defect budget, and persistence window.
$\mathcal{P}_O$	Family of observer-level predicates used in the specification.
ε_O	Defect budget allowed by the specification.
τ_O	Persistence window required by the specification.
v^*	Occupied-volume completion threshold.
q^*	Lattice-order completion threshold.
$\varepsilon_{\mathrm{seed}}$	Admissible seed defect tolerance.
$\tau_{\mathrm{life}}(X)$	Observer-level persistence surrogate of the realized object.
R_{pre}	Precursor reservoir available for materialization.
$R_{\mathrm{pre}}^{\mathrm{cryst}}$	Crystalline precursor reservoir admissible for lattice-forming many-body phases.
$R_{\mathrm{pre}}^{(a)}$	Monatomic precursor reservoir for a single allowed species a.
Γ	Assembly history transforming precursor sectors into a candidate target sector.
$\mathfrak{C}$	Controlled assembly channel, that is, a family of admissible assembly histories.
$\mathfrak{C}_{\mathrm{cryst}}$	Controlled crystallization channel.
S_{seed}	Seed sector carrying the local composition and lattice regime needed for ordered growth.
$F(X_n)$	Growth front of a partially assembled object at growth step n.
X	Candidate realized sector produced by the channel.
$D(X)$	Defect sector of a realized crystalline object relative to its ideal target.
$\lvert D(X)\rvert_O$	Observer-weighted defect norm.
ξ_n	Stochastic defect injection at growth step n.
J_n	Front mismatch observable at growth step n.
λ	Retained fraction of inherited defect load.
η, ζ	Sensitivity coefficients in the defect recurrence estimate.
$\mathcal{C}_O(X)$	Completion functional collecting the observer-level occupied-volume fraction, lattice-order quality, and defect norm.
N_{comp}	Completion step, that is, the first growth index at which the completion criteria are satisfied.
$\mathcal{H}_0, \ldots, \mathcal{H}_8$	Hierarchy of realized-object classes from simple hard body to self-fabricating system.

Canonical Theory Ledger

Item	Status	Canonical interpretation
M0	[A]	Boundary principle: no primitive materialized object and no ex nihilo reading; all claims are relative to precursor resources and admissible assembly histories.
M1–M6, M8–M13	[A]	Specification, precursor, channel, soft/hard success, and observer-equivalence definitions form the canonical first language of the program.
M7	[Open-Calibration]	Resource-profile bookkeeping is retained only as a later extension and is not used in Parts I–XI.
M-A1	[A]	No ex nihilo materialization axiom rendered in the present manuscript pass.
M-A2–M-A6	[Open-Closure]	Additional axiomatic package retained in the legacy source corpus and not yet normalized into the present LaTeX pass.
M-P1–M-P3	[D]	Specification relativity, materialization stronger than synthesis, and informational sectors as privileged precursors.
M-T1–M-T4	[T]	No ex nihilo, soft/hard separation, observer-relative identity, and minimal precursor target theorems.
M-F1–M-F5	[F]	Mandatory audited failure modes for instability, invisibility, nonpersistence, defect overflow, and missing carrier class.
M-MP1	[Internal-M]	First model problem fixed as hard materialization of a simple crystalline body.
M14–M26	[A]	Crystalline specification, defect calculus, crystalline channels, crystalline success, and soft/hard crystalline split.
M-C1	[T]	First crystalline hard-materialization target theorem.
C-F1–C-F6	[F]	Crystalline failure modes for reservoir insufficiency, instability, glass capture, defect overflow, invisibility, and nonpersistence.
M27–M40	[A]	Monatomic target, seed, growth-history, completion-functional, defect, nucleation, and completion-time definitions.
M-A7	[A]	Controlled defect recurrence axiom for ordered monatomic growth.
M-P4–M-P6	[D]	Seed barrier-lowering, bounded-defect growth, and finite completion propositions.
M-C2–M-C3	[T]	Seeded completion and nucleation-reduction target theorems for the monatomic regime.
M-F6–M-F10	[F]	Monatomic failure modes for missing seeds, front disordering, defect runaway, glass capture, and threshold incompletion.
M-A1-Mono	[A]	First computational schema of the monatomic regime; the manuscript disambiguates a locally reused source label.

Hierarchy Ledger

Level	Status	Normalized description
$\mathcal{H}_0$	[Internal-M] / [Open-Calibration]	Hard materialization of a simple crystalline body; first level with an explicit lower-regime manuscript draft and a possible later engineering bridge.
$\mathcal{H}_1$	[Open-Calibration]	Usable body with theorem architecture present but calibration still open.
$\mathcal{H}_2$	[Open-Closure]	Simple composite device.
$\mathcal{H}_3$	[Open-Closure]	Stateful device with memory-bearing sectors.
$\mathcal{H}_4$	[Open-Closure]	Adaptive device with bounded correction and recovery channels.
$\mathcal{H}_5$	[Internal→Observer] / [Open-Closure]	Strongly adaptive device with informational closure.
$\mathcal{H}_6$	[Open-Closure]	Proto-agent object with symbolic-state control.
$\mathcal{H}_7$	[Open-Closure]	Self-extending fabricator.
$\mathcal{H}_8$	[Open-Closure]	Self-fabricating system with heredity and recursive continuation.

Theorem and Formal-Support Provenance Table

Item	Status	Support class	Comment
M0	[A]	Source-faithful boundary principle	Rules out primitive-object and ex nihilo readings at the manuscript boundary.
M1–M6, M8–M13	[A]	Source-faithful program definitions	Presently rendered as manuscript primitives rather than fully closed formal derivations.
M7	[Open-Calibration]	Deferred bookkeeping remark	Resource-profile bookkeeping is retained only as a later extension and is not used in the present manuscript pass.
M-A1	[A]	Source-faithful program axiom	Explicitly rendered no-ex-nihilo rule for admissible precursor-based claims.
M-P1–M-P3	[D]	Derived from imported book architecture	Treated as consequences expected from the observer and chemistry/material layers plus new definitions.
M-T1	[T]	Target theorem	Requires theorem-chain closure from precursor existence and admissible histories.
M-T2	[T]	Target theorem	Separates externally supported realizations from persistent hard realizations.
M-T3	[T]	Target theorem	States observer-relative identity rather than microscopic uniqueness.
M-T4	[T]	Target theorem	Opens the minimal-sufficient-reservoir program needed for future fabricator theory.
M-F1–M-F5	[F]	Audited failure basis	Failure modes are normalized as mandatory rigor conditions, not as optional commentary.
M-MP1	[Internal-M]	Model-problem fixation	Commits the next strict specialization to the simple crystalline hard-materialization regime.
M14–M26	[A]	Source-faithful crystalline definitions	Presently rendered in manuscript form and not claimed here as full closure theorems.
M-C1	[T]	Crystalline target theorem	First hard-materialization theorem in the crystalline regime; currently a target theorem with dependency note only.
C-F1–C-F6	[F]	Crystalline audited failure basis	Extends the general failure vocabulary to the first explicit crystalline object class.
M27–M40	[A]	Source-faithful monatomic definitions	Monatomic targets, seed sectors, growth histories, completion observables, defect recurrence, seedless nucleation, and completion time.
M-A7	[A]	Source-faithful monatomic axiom	Controlled defect recurrence for the ordered-growth regime.
M-P4–M-P6	[D]	Derived monatomic	Barrier lowering by seeds, bounded-

Section Ledger

Current chapter	Source block	Normalization status
Protocol, Boundary, and Status Policy	Part I, Sections 1–4	Source-faithful summary with explicit status discipline and manuscript boundary.
Imported Foundations and Program Boundary	Part I, Sections 1–3	Imported structural, observer, composite, and material-class dependencies rendered in clean English prose.
Specification Theory	Part II	Definitions M1–M5 normalized into theorem-style LaTeX.
Precursor and Resource Theory	Part III	Definition M6, Axiom M-A1, and Definition M7 rendered in canonical form.
Assembly Histories and Controlled Channels	Part IV	Definitions M8–M10 rendered in canonical form.
Success Criteria	Part V	Definitions M11–M13 rendered in canonical form.
Hierarchy Placement and First Derived Propositions	Part VI plus internal hierarchy ledger	Source-faithful propositions with explicit note that the fuller hierarchy remains an internal program ledger.
First Target Theorems	Part VII	Target Theorems M-T1–M-T4 rendered without closure claims.
Failure Modes, Provenance, and the First Model Problem	Part VIII and Part IX	Failure modes M-F1–M-F5 and the first crystalline model problem normalized in manuscript form.
Engineering Transition and Next Dependencies	Editorial bridge after Part IX	Records the downstream continuation path after the now-rendered crystalline specialization.
Crystalline Target Objects	Part X	Source-faithful crystalline specification, defect calculus, channel definitions, first crystalline target theorem, and crystalline failure regime.
Monatomic Crystalline Fabrication	Part XI	Source-faithful monatomic target, seed sector, growth history, completion functional, bounded-defect criterion, and monatomic completion theorems.

B Downstream P0 Computational Provenance (Non-Evidential)

This appendix records downstream computational provenance only as non-evidential context. None of the items below is used in the core manuscript as theorem evidence.

Engineering Bridge Note

Within the internal program ledger, $\mathcal{H}_0$ is the first level for which a later engineering bridge can be formulated. A downstream P0 computational line may track threshold observables, crystalline quality surrogates, metastability windows, growth surrogates, and run-level verdict classes for the lowest hard-materialization regime. These entries remain downstream provenance rather than evidence for the target-theorem chain in the core manuscript.

P0 Computational Verdicts

Verdict	Normalized meaning
FAIL_RESERVOIR	Materialization attempt fails before growth because precursor sufficiency is not met.
FAIL_GROWTH	Growth never reaches a target-compatible completed state.
SOFT_SUCCESS	Observer-level completion occurs in the downstream prototype, but the prototype-level hard-persistence criteria are not certified.
GLASS_CAPTURE	Completion is trapped in a glass-like or metastable capture regime distinct from hard success.
HARD_SUCCESS	Prototype-side verdict label used when the operational surrogates for hard completion and metastable persistence are jointly reported as satisfied.

P0 Calibration Note. A numerical prototype calibration snapshot is omitted here, because downstream raw prototype records are not used in the core manuscript either as theorem evidence or as canonical calibration data.

C Source Corpus and Provenance

The following items are source-corpus and provenance records for the present source-faithful reconstruction. They are not presented as an external scholarly bibliography.

References

[1] Catman. *Main Structural-Physical Theory Manuscript*. Working repository manuscript, 2026. Repository path: `docs/book/main.tex`. Snapshot date: 2026-04-14. Internal document version ID: `docs/book/main.tex`@2026-04-14. Repository commit: unavailable in current local snapshot metadata. Access date: 2026-04-14. Access status: internal workspace source.

[2] Catman. *Matria_v1*. Materialization primary source manuscript, version v1, 2026. Repository path: `docs/Technology/Matria_v1.md`. Snapshot date: 2026-04-14. Internal document version ID: `Matria_v1.md`:v1. Repository commit: unavailable in current local snapshot metadata. Access date: 2026-04-14. Access status: internal workspace source.

[3] Catman. *Matria_Dev_v1*. Materialization development corpus, version v1, 2026. Repository path: `docs/Technology/Matria_Dev_v1.md`. Snapshot date: 2026-04-14. Internal document version ID: `Matria_Dev_v1.md`:v1. Repository commit: unavailable in current local snapshot metadata. Access date: 2026-04-14. Access status: internal workspace source.

[4] Catman. *Materialization Program README*. Canonical workspace guide, 2026. Repository path: `docs/Technology/programs/materia/README.md`. Snapshot date: 2026-04-14. Internal document version ID: `programs/materia/README.md`@2026-04-14. Repository commit: unavailable in current local snapshot metadata. Access date: 2026-04-14. Access status: internal workspace source.